Sub-Task 1

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Contents

1 Experiment Report

1.1 Methodology

The ranking task was formulated as a token classification problem where each element in a sequence predicts its own rank. Three baseline models (MLP, RNN and LSTM) were implemented and compared against a Transformer Encoder built from scratch.

The Transformer implementation consisted of:

- Positional Encoding
- Multi-Head Self Attention
- Feed Forward Network
- Residual Connections
- Layer Normalization
- Classification Head

Each model received a sequence of numerical values and predicted the rank of every element in the sequence. Cross Entropy Loss was used for training and token-level accuracy was used for evaluation.

1.2 Architectural Choices

The final Transformer architecture consisted of:

- Input Projection: $1 \rightarrow 64$
- Positional Encoding
- Two Transformer Encoder Blocks
- Four Attention Heads
- Feed Forward Dimension: 128
- Classification Layer: $64 \rightarrow 10$

The model treats ranking as a token classification task where each token predicts one of ten possible ranks.

1.3 Numerical Representation Strategies

Several numerical representation methods were investigated:

1. Raw numerical inputs
2. Z-score normalization
3. Min-Max scaling
4. Sequence-wise normalization
5. Embedding-based representation

The objective was to determine whether the Transformer benefits more from preserving numerical relationships or treating values as independent symbols.

1.4 Baseline Model Comparison

Model	Test Accuracy
MLP	46.84%
RNN	43.08%
LSTM	43.98%
Transformer	95.93%

Table 1: Baseline Model Comparison

The Transformer significantly outperformed all baseline models. While MLP, RNN and LSTM struggled to learn accurate ranking relationships, the Transformer was able to effectively compare elements across the sequence using self-attention.

1.5 Attention Visualization

Attention heatmaps were generated from the second encoder layer after training to understand how the Transformer performs ranking.

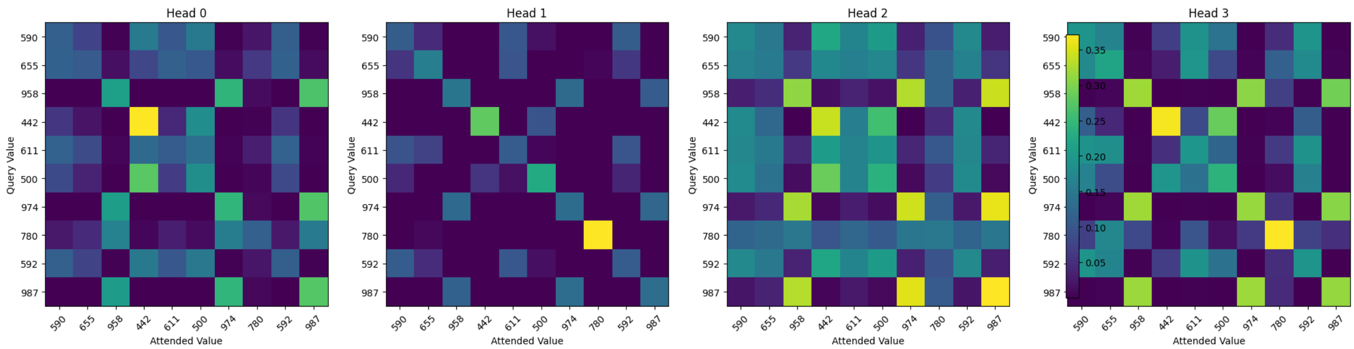

Figure 1: Attention heatmaps from the four attention heads in the second Transformer encoder layer. Rows represent query values and columns represent attended values.

1.5.1 Attention Analysis

Figure 1 shows the attention patterns learned by the four attention heads in the second encoder layer.

Head 0 distributes attention across multiple values in the sequence. Larger values such as 958, 974 and 987 receive relatively high attention, suggesting that these values act as reference points during ranking.

Head 1 exhibits highly concentrated attention patterns. Each query value attends strongly to only a few specific values, indicating that this head performs targeted comparisons between selected elements.

Heads 2 and 3 display more structured behavior. These heads consistently assign higher attention to larger values while giving lower attention to smaller values. Values such as 958, 974 and 987 frequently receive strong attention across many queries.

The heatmaps also show evidence of pairwise comparisons. Smaller values often attend to larger values, while larger values attend to other high-valued elements. This indicates that rank prediction relies on relationships between multiple elements rather than independent processing.

Different heads specialize in different comparison strategies. Some heads focus on broad comparisons across the sequence, while others focus on specific values. This specialization highlights one of the main advantages of multi-head attention.

Overall, the attention visualizations indicate that the Transformer learns ranking by comparing values against one another rather than memorizing positions or processing each value independently.

1.6 Out-of-Distribution Testing

The model was evaluated on sequences that differed significantly from the training distribution.

1.6.1 Results

Sample 1

Input:

$$[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]$$

Predicted Ranks:

$$[0, 1, 1, 2, 6, 6, 7, 8, 8, 9]$$

Sample 2

Input:

$$[10, 9, 8, 7, 6, 5, 4, 3, 2, 1]$$

Predicted Ranks:

$$[9, 8, 7, 6, 6, 6, 6, 6, 1, 0]$$

Sample 3

Input:

$$[1000, 900, 800, 700, 600, 500, 400, 300, 200, 100]$$

Predicted Ranks:

$$[9, 8, 7, 6, 5, 4, 3, 2, 1, 0]$$

Sample 4

Input:

$$[5, 5, 5, 5, 5, 5, 5, 5, 5, 5]$$

Predicted Ranks:

$$[9, 9, 9, 9, 9, 9, 9, 9, 9, 9]$$

Sample 5
Input:

$$[1, 1000, 2, 999, 3, 998, 4, 997, 5, 996]$$

Predicted Ranks:

$$[0, 7, 0, 7, 0, 7, 2, 7, 2, 7]$$

1.6.2 Analysis

The Transformer performed well on inputs containing values much larger than those seen during training. The sequence containing values from 1000 to 100 was ranked correctly, indicating that the model learned relative comparisons rather than memorizing specific numerical ranges.

The model struggled on perfectly sorted sequences and highly structured patterns. Several ranks were repeated in the ascending sequence, suggesting that the model was unable to produce a perfectly ordered ranking despite the simple pattern.

For the identical-value sequence, all elements were assigned the same rank. Since there is no clear ordering among identical values, this behavior is reasonable.

The alternating high-low sequence showed that the model could distinguish between large and small values but struggled to assign precise ranks to every element.

Overall, the Transformer demonstrated strong generalization beyond the training distribution while still exhibiting weaknesses on highly structured and ambiguous inputs.

1.7 Ablation Studies

1.7.1 Effect of Number of Attention Heads

Heads	Test Accuracy
4	97.31%
8	97.38%

Table 2: Effect of Number of Attention Heads

Increasing the number of attention heads provided only a marginal improvement in performance. The small gain suggests that four heads are already sufficient to capture the relationships required for ranking.

1.7.2 Effect of Learning Rate

Reducing the learning rate from 10^{-3} to 5×10^{-4} improved the test accuracy from 95.93% to 97.34%.

Training also became more stable, with fewer fluctuations in validation accuracy. This experiment shows that optimization settings can have a significant impact on Transformer performance and convergence.

1.7.3 Positional Encoding Ablation

Configuration	Test Accuracy
With Positional Encoding	97.34%
Without Positional Encoding	73.36%

Table 3: Effect of Positional Encoding

Removing positional encoding resulted in a significant drop in performance. Although ranking primarily depends on numerical comparisons, positional information helps the Transformer learn more reliable relationships between sequence elements. The model also exhibited less stable training when positional encoding was removed.

1.7.4 Continuous Representation vs Embedding Representation

Representation	Test Accuracy
Continuous Projection	97.34%
Embedding Layer	80.25%

Table 4: Continuous vs Embedding Representation

The continuous representation substantially outperformed the embedding-based representation. This indicates that preserving numerical relationships between values is important for the ranking task. Treating numbers as independent symbols makes it more difficult for the model to learn ordering relationships.

1.7.5 Effect of Normalization

Representation	Test Accuracy
Raw Input	97.34%
Z-Score	97.20%
Min-Max	97.80%
Sequence-Wise	98.34%

Table 5: Effect of Different Normalization Methods

Sequence-wise normalization achieved the best performance among all normalization methods tested.

Z-score normalization produced performance very similar to raw inputs, while Min-Max scaling resulted in a small improvement. Sequence-wise normalization performed best because ranking depends on relative ordering rather than absolute magnitude. By removing scale information and preserving only the relationships within each sequence, the model was able to learn the ranking task more effectively.

1.7.6 Effect of Transformer Depth

Layers	Test Accuracy
1	50.74%
2	95.17%

Table 6: Effect of Transformer Depth

Reducing the model to a single encoder layer caused a substantial decrease in performance. The one-layer model struggled to learn meaningful ranking relationships, whereas the two-layer model converged successfully.

This result suggests that multiple encoder layers are important because they allow repeated refinement of attention-based representations before final rank prediction.

1.8 Evaluation Metrics

The primary evaluation metric used throughout all experiments was token-level accuracy.

Token-level accuracy measures the percentage of sequence elements for which the predicted rank exactly matches the ground-truth rank. Since the ranking task requires predicting a rank for every element in the sequence, this metric provides a direct measure of model performance.

Training loss was also monitored using Cross Entropy Loss to evaluate convergence and training stability.

1.9 Conclusions

The Transformer consistently outperformed the MLP, RNN and LSTM baselines on the ranking task. Self-attention enabled the model to compare elements across the entire sequence and learn complex ranking relationships more effectively than recurrent or feed-forward architectures.

Several important observations emerged from the experiments:

- Positional Encoding significantly improved performance and training stability.
- Continuous numerical representations were substantially more effective than embedding-based representations.
- Sequence-wise normalization achieved the highest overall accuracy of 98.34%.
- Increasing the number of attention heads provided only marginal improvements.

- Learning rate had a noticeable impact on convergence and final performance.
- Multiple encoder layers were important for learning accurate ranking relationships.
- Attention visualizations showed that the model learns ranking through interactions between elements rather than independent processing.
- Out-of-distribution testing demonstrated strong generalization to unseen value ranges, while revealing weaknesses on highly structured patterns.

Overall, the experiments demonstrate that Transformer-based models are highly effective for ranking tasks when numerical relationships are preserved and appropriate input representations are used. The results further show that representation choices and normalization strategies can influence performance as much as architectural modifications.